Unsupervised Learning

Unsupervised Learning

Goals: Search for hidden patterns or groupings in a dataset. Analyze/cluster unlabeled datasets.

Has some similarities to exploratory data analysis (EDA).

Types of Unsupervised Learning

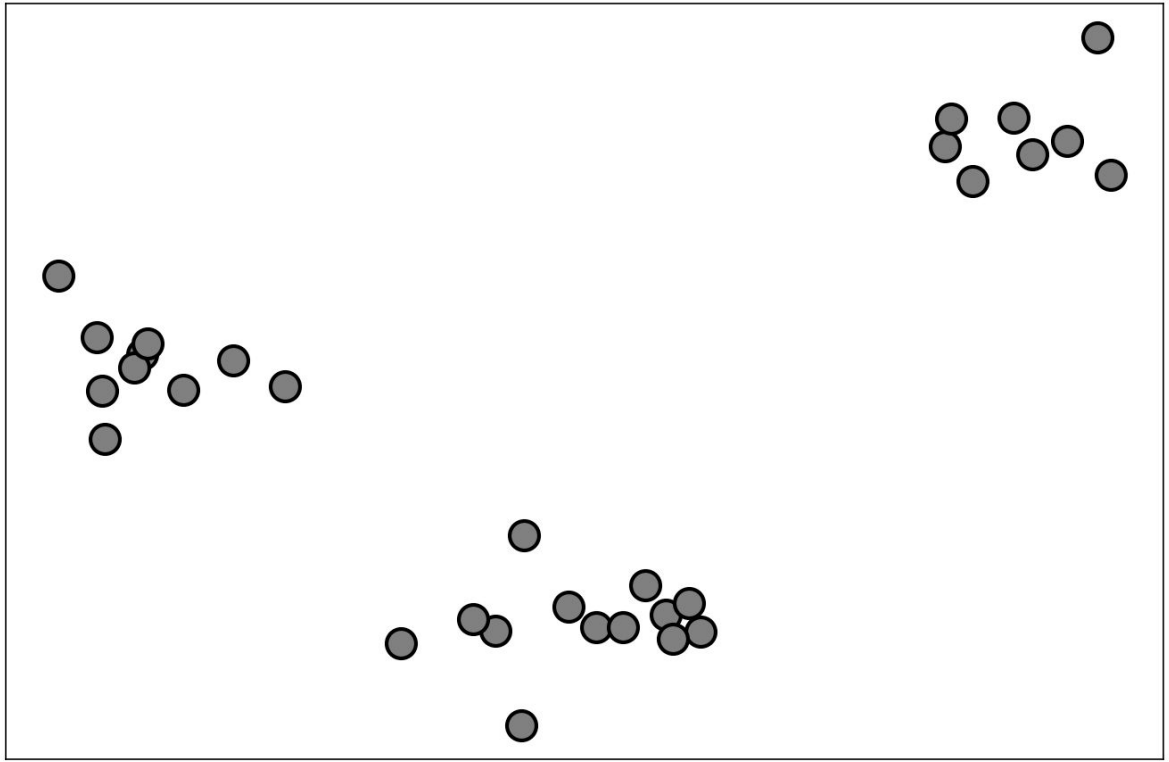
Four main goals:

- **Clustering:** discover “clumps” of similar points (eg. customer segmentation)
- **Embedding:** Find a lower-dimensional representation of the data, without losing too much “resolution”
- **Density Estimation:** Approximate the probability distribution of the data (think KDE from seaborn). Can also search for anomalies.
- Finding good explanations (**hidden causes** or **sources**) of the data (model the data generating process)

Unsupervised Learning Challenges

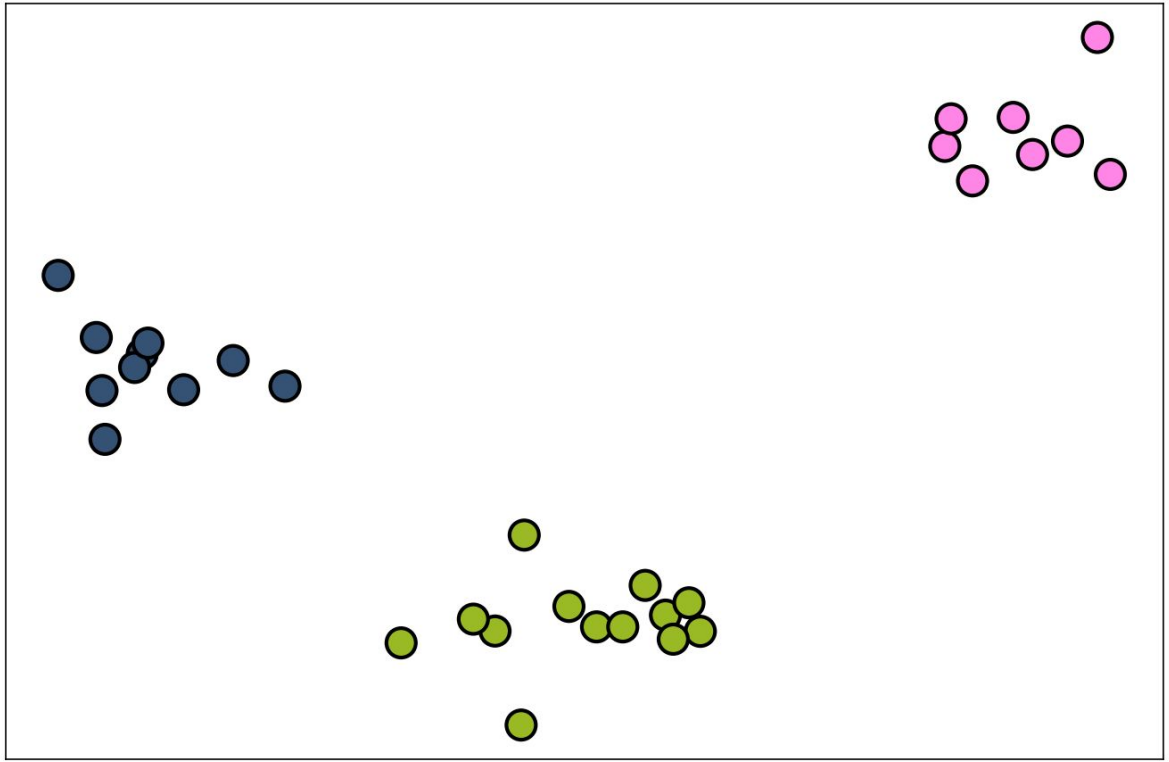
- No simple goal, like with supervised learning.
- No teacher/supervisor to provide “correct answers”.
- Hard to assess the results obtained - no labels, so we can’t compute accuracy or MAE.

Unsupervised Learning - Clustering



Can be used to uncover groupings or clusters in a dataset.

Unsupervised Learning - Clustering



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Unsupervised Learning - Dimensionality Reduction

Can give a lower-dimensional representation of a high-dimensional dataset.

	label	A_23_P100001	A_23_P100011	A_23_P10002
0	control	-0.3259	-0.1298	-0.179
1	control	-0.2818	-0.2255	0.045
2	control	-0.2518	-0.1426	-0.033
3	control	-0.3019	-0.1793	0.464
4	control	-0.3986	-0.1259	-0.299

5 rows × 25074 columns

25,073-Dimensional Dataset

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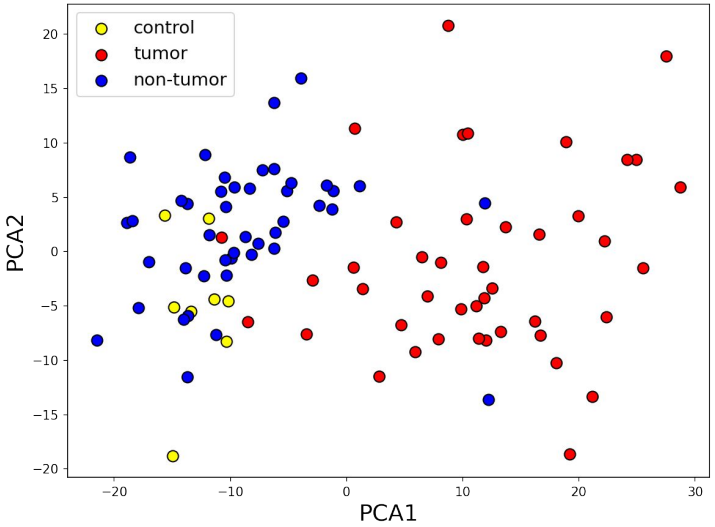


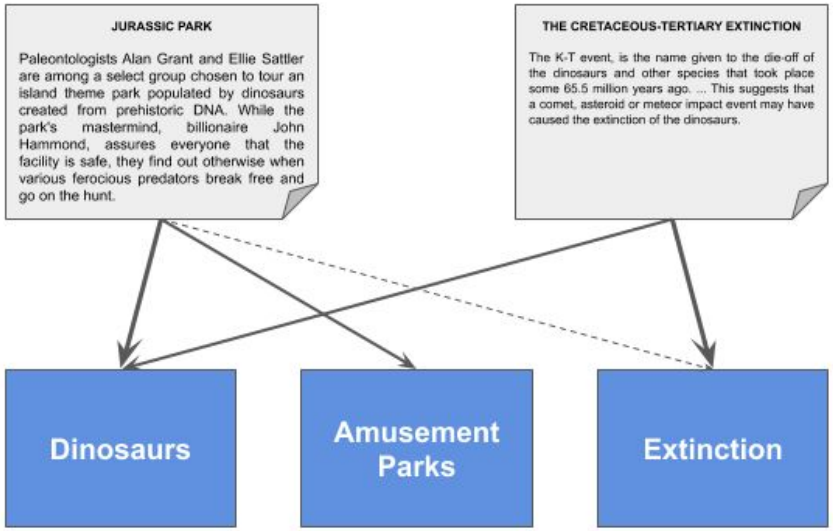
25,073-Dimensional Dataset

2-Dimensional Representation

Unsupervised Learning - Topic Modeling

Cluster word groups and similar expressions that best characterize a set of documents.

We'll see this more when we look at NLP.



https://www.baeldung.com/cs/latent-dirichlet-allocation